

MOHAMMAD SOHAIL AHMED

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Hyderabad, India

PROFESSIONAL SUMMARY

Backend-focused Software Engineer with 3+ years of experience building scalable microservices using Java, Spring Boot, and AWS. Experienced in designing high-performance REST APIs, event-driven architectures, and cloud-native systems for large-scale commerce platforms (Changi Airport Group). Strong in production support, system observability (Dynatrace, Datadog), and performance optimization. Also experienced in AI-driven backend systems including RAG architectures, vector databases, and LLM-based applications.

PROFESSIONAL EXPERIENCE

Software Engineer | CJSS Technologies Pvt. Ltd., Hyderabad

January 2024 – Present

Client: Changi Airport Group (CAG)

Project: TREX – Scalable Microservices Commerce Platform for iShopChangi

- Played a key role in developing **TREX**, a scalable microservices-based commerce management platform powering **iShopChangi**, enabling seamless handling of products, pricing, orders, merchants, users, and promotions.
- Built and optimized **RESTful APIs and event-driven workflows** using AWS Lambda and AppSync, improving system responsiveness and supporting high-traffic retail operations.
- Integrated enterprise systems such as **SAP and Okta** to ensure secure authentication, authorization, and seamless business process synchronization.
- Deployed containerized applications using Docker and managed them on AWS ECS, ensuring high availability, scalability, and reliability of services.
- Implemented advanced **observability and monitoring solutions** using **Dynatrace** and **Datadog** for distributed tracing, log analysis, and performance optimization.
- Led and supported **production deployments, incident troubleshooting, and performance tuning**, ensuring adherence to SLAs and minimizing downtime.
- Collaborated with cross-functional teams in an **Agile environment**, delivering high-quality features aligned with business and client requirements.

TECHNICAL SKILLS

Languages:

Java, Python, C, SQL, JavaScript

Backend & Distributed Systems:

Spring Boot, Spring MVC, Hibernate, JPA, REST APIs, Microservices, System Design, Event-Driven Architecture, Scalable Backend Systems

Cloud & Infrastructure:

AWS (EC2, S3, Lambda, AppSync, CloudWatch, ELB, Auto Scaling), Cloud-Native Architecture, Serverless Computing, High Availability Systems, Distributed Systems

DevOps & CI/CD:

Docker, Jenkins, CI/CD Pipelines, Deployment Automation

Databases:

MySQL, DynamoDB, Relational & NoSQL Databases, Data Modelling, Query Optimization

Frontend:

ReactJS, JavaScript, HTML5, CSS3

Testing & Quality:

JUnit, Mockito, Unit & Integration Testing, Test-Driven Development (TDD)

Observability & Reliability:

Dynatrace, Datadog, Monitoring & Logging, Distributed Tracing, Performance Monitoring, SLA/SLO Management

AI / LLM Systems:

RAG (Retrieval-Augmented Generation), Multi-Agent Systems, Vector Databases (Qdrant), Embeddings (Gemini), Semantic Search, LLM Orchestration, Context-Aware AI Systems, Information Retrieval, Metadata Filtering

Engineering Practices:

Agile Methodology, API Design, Production Debugging, Performance Optimization, Incident Management, Scalable System Design

PROJECTS

AI Chatbot with Multi-Agent Orchestration using RAG (Gemini + Qdrant)

- Designed and implemented an end-to-end **multi-agent RAG system** using **Spring Boot, Python FastAPI, Gemini AI, and Qdrant vector database**.
- Built a **microservices-based architecture** with specialized agents for ingestion, embedding generation, retrieval, and response synthesis.
- Implemented a **document ingestion agent** for file upload, Base64 decoding, text extraction, chunking, and metadata enrichment.
- Integrated a **Gemini embedding agent** to generate vector embeddings and stored them in **Qdrant with payload-based filtering for role/group access control**.
- Built a **retrieval agent** enabling secure, role-aware semantic search using vector similarity and metadata constraints.
- Developed a **Gemini LLM-based generation agent** to produce responses grounded strictly in retrieved context (RAG-based answers).
- Improved retrieval accuracy using **vector indexing, filtering, and structured query constraints in Qdrant**.

CONNECT – Enterprise Workforce & Business Operations Platform

- Designed and developed **CONNECT**, a full-stack enterprise platform for HR operations, recruitment, vendor management, asset tracking, and employee learning.
- Built scalable Spring Boot microservices architecture supporting employee lifecycle, hiring workflows, vendor invoices, timesheets and asset management.
- Integrated Microsoft Authentication (Azure AD) for secure enterprise login and centralized identity management.
- Implemented role-based access control (RBAC) ensuring secure and role-specific access across HR, finance, managers, and employees.
- Built modules for vendor invoice processing and asset management, improving enterprise operational efficiency.
- Designed a gamified learning platform to enhance employee engagement and structured skill development.
- Optimized MySQL relational schemas for scalable enterprise data management and reporting.

EDUCATION

Bachelor of Technology (Computer Science and Engineering)

Rajiv Gandhi University of Knowledge Technologies, Basar

Pursued degree alongside full-time professional experience in software engineering

May 2024 | GPA: 8.9/10

CERTIFICATIONS

- Oracle Certified Associate (OCA), Java SE 8 Programmer
- AWS Certification – Abhitrainings.org
- Blockchain and Its Applications – NPTEL

PERSONAL DETAILS

- Nationality: Indian
- Date of Birth: 5 April 2002
- Visa Status: Open to Relocation / Visit Visa
- Languages: English, Urdu, Hindi, Telugu